

Skin, Soft Tissue, Bone, and Joint Infections

Medvale Internal Medicine - Infectious Diseases

Why this matters clinically

Skin, soft tissue, bone, and joint infections sit at the intersection of routine outpatient medicine and true source-control emergencies. A small abscess may need drainage more than broad antibiotics, while a patient with pain out of proportion, crepitus, shock, or rapidly advancing erythema may need immediate surgical exploration for necrotizing infection. Bone and joint infections require a different mental model because antibiotics must penetrate poorly vascularized tissue, diagnostic cultures are often obtained invasively, and delayed drainage can leave permanent structural damage.

This chapter focuses on cellulitis, erysipelas, purulent abscess, necrotizing soft tissue infection, lymphadenitis and lymphangitis, tetanus wound management, diabetic foot infection, osteomyelitis, septic arthritis, and prosthetic or hardware-associated infection. Cardiovascular infections, zoonotic and travel infections, fungal and parasitic disease, antimicrobial pharmacology, prophylaxis, vaccines, and infection control will be separated into later infectious disease segments.

Clinical anchor: if the patient is toxic, hypotensive, immunocompromised, has rapidly progressive pain or swelling, has a diabetic foot wound with systemic signs, has suspected septic arthritis, or has possible necrotizing infection, treat the situation as a source-control emergency rather than a simple outpatient antibiotic choice.

Core illness scripts

Nonpurulent cellulitis. Diffuse erythema, warmth, tenderness, and edema without abscess or drainage. Fever may be absent. Most cases are driven by beta-hemolytic streptococci, with MSSA possible depending on portal of entry. The decisive action is to assess severity, portal of entry, MRSA risk, systemic toxicity, and need for IV therapy.

Purulent abscess. Fluctuant nodule or collection with possible drainage and surrounding erythema. *Staphylococcus aureus*, including MRSA in the right epidemiologic setting, is the major organism to consider. Incision and drainage is the key intervention. Antibiotics are added for systemic signs, extensive disease, immunocompromise, difficult anatomic sites, recurrent infection, or failure of drainage alone.

Necrotizing soft tissue infection. Severe pain out of proportion, rapidly progressive edema or erythema, bullae, skin anesthesia, crepitus, shock, or organ dysfunction. The microbiology may be polymicrobial or monomicrobial, including group A *Streptococcus*, *Clostridium*, *Vibrio*, or *Aeromonas* depending on exposure. The decisive action is immediate surgical consultation and exploration plus broad IV antibiotics. Imaging should not delay surgery in an unstable patient.

Diabetic foot infection. Ulcer with purulence or local inflammation in a patient with neuropathy, ischemia, or impaired wound healing. Early mild infections are often gram-positive; chronic, deep, ischemic, necrotic, or previously treated wounds may be polymicrobial with gram-negatives and anaerobes. The decisive action is to classify severity, evaluate vascular supply and bone involvement, debride or drain when needed, culture deep tissue when possible, and offload pressure.

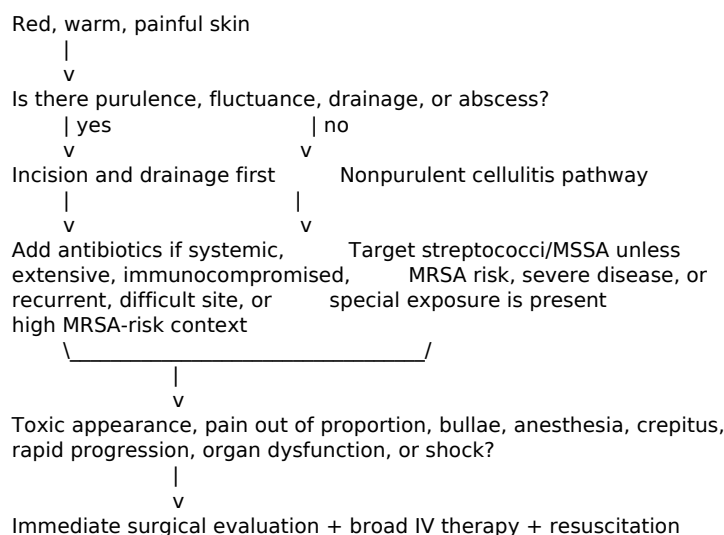
Osteomyelitis. Infection-driven inflammation and destruction of bone. Fever may be absent. Focal bone pain, chronic ulcer over bone, sinus tract, or persistent inflammatory markers should raise suspicion. *S. aureus* is the most common organism overall, but route and host determine the differential. MRI and culture-directed therapy are central; bone biopsy is valuable when diagnosis or microbiology is uncertain.

Septic arthritis. Acutely hot, swollen, painful joint with markedly limited active and passive range of motion. *S. aureus* is the most common adult cause. Gonococcal disease should be considered in sexually active young adults; gram-negative rods are more likely in older, immunocompromised, or healthcare-associated patients. Urgent arthrocentesis, cultures, antibiotics, and drainage are the key actions.

A practical classification framework

The first split is anatomic depth. Epidermal and superficial dermal infections behave differently from infections that cross into fascia, muscle, joint space, or bone. The second split is purulence. Pus implies a collection or staphylococcal phenotype and raises the value of drainage; nonpurulent diffuse erythema is more often streptococcal. The third split is host and exposure. Diabetes, vascular insufficiency, injection drug use, water exposure, animal bites, recent surgery, hardware, immunosuppression, and neutropenia all widen the microbiologic differential. The fourth split is physiologic severity: hypotension, tachycardia, altered mentation, rising lactate, acute kidney injury, rapidly worsening pain, and extensive tissue injury should shift the plan from outpatient treatment to urgent inpatient therapy and source control.

Original triage diagram:



Cellulitis

Cellulitis is infection of the dermis and subcutaneous tissue that produces poorly demarcated erythema, warmth, swelling, and tenderness. It is usually diagnosed clinically. A portal of entry may be obvious, such as a wound, ulcer, catheter site, bite, surgical incision, or injection site, but it may also be subtle, such as toe-web fissures from tinea pedis, eczema, venous stasis dermatitis, chronic edema, or minor trauma. The uploaded source emphasizes that treatment should be based on suspected pathogens from the patient history and that untreated cellulitis can progress to bacteremia or life-threatening deep infection.

Classic nonpurulent cellulitis is usually streptococcal. *Staphylococcus aureus* becomes more likely when cellulitis is associated with penetrating trauma, abscess, drainage, injection drug use, prior MRSA, or purulence. Recurrent cellulitis often reflects persistent edema, lymphatic injury, venous disease, obesity, toe-web fungal infection, or skin barrier breakdown. Treating the acute infection without correcting edema or skin breakdown leads to repeated episodes.

Cellulitis versus common mimics

Uncomplicated cellulitis usually has diffuse warmth and erythema, tenderness proportional to the exam, and a possible portal of entry. Fever may be absent. The diagnosis is usually clinical.

Abscess or purulent infection has a focal fluctuant or draining center. The key action is drainage. Antibiotics alone often fail because the antibiotic cannot sterilize a walled-off collection.

DVT can mimic cellulitis with a swollen, warm, tender leg. It is more likely when swelling is deep, calf tenderness predominates, erythema is less inflammatory, venous risk factors are present, or the diagnosis is uncertain. Compression ultrasound is appropriate when concern is meaningful.

Venous stasis dermatitis and lymphedema often produce bilateral erythema, scaling, hyperpigmentation, chronic edema, and relatively little pain. Bilateral red legs should not automatically be labeled cellulitis.

Necrotizing infection should be suspected with pain out of proportion, rapid spread, bullae, skin anesthesia, crepitus, shock, or organ dysfunction. Do not rely on superficial skin appearance alone.

Diagnosis and management of cellulitis

Routine blood cultures have low yield in uncomplicated cellulitis. Cultures are more useful when the patient is systemically ill, immunocompromised, neutropenic, has immersion injury, animal bite, recurrent severe infection, or unusual exposure. Imaging is not needed for most uncomplicated cases, but ultrasound can detect occult abscess and MRI or CT may be needed when deeper infection, foreign body, gas, osteomyelitis, or septic thrombophlebitis is suspected.

Treatment begins with severity classification. Mild nonpurulent cellulitis can often be treated orally with an agent active against streptococci and MSSA. MRSA-active coverage is added when there is purulence, penetrating trauma, prior MRSA, injection drug use, or severe infection. Hospitalize patients with systemic toxicity, rapidly progressive infection, failure of outpatient therapy, inability to take oral medication, severe immunocompromise, concern for necrotizing infection, or major comorbid risk. Elevation of the limb, edema control, wound care, and treatment of tinea pedis or ulcers are part of recurrence prevention.

Pitfall: worsening erythema during the first day of therapy does not always mean failure, because inflammatory changes can lag behind bacterial killing. Worsening pain, systemic signs, new bullae, necrosis, hypotension, or rising lactate are different; those findings require urgent reassessment for deep infection or wrong diagnosis.

Erysipelas

Erysipelas is a more superficial cellulitis of the upper dermis and superficial lymphatics. It tends to produce a bright red, raised, sharply demarcated, tender plaque and is classically caused by group A *Streptococcus*. It is often seen on the lower extremities or face and may occur in patients with lymphatic obstruction, venous disease, trauma, fungal infection, alcohol use disorder, diabetes, or prior surgery.

Because erysipelas is more superficial, its border may look more dramatic than deeper cellulitis. Uncomplicated disease is treated with streptococcal-active therapy. If abscess, drainage, penetrating trauma, prior MRSA, or severe systemic illness is present, manage it as a broader cellulitis or purulent infection rather than as isolated erysipelas. Recurrence is common when lymphatic damage or edema persists.

Purulent skin infection and abscess

A purulent infection is defined by pus: abscess, furuncle, carbuncle, draining wound, or fluctuant collection. The most important intervention is source control. Antibiotics penetrate poorly into a walled-off abscess, so incision and drainage is often more important than the specific oral antibiotic. Culture is most useful when infection is severe, recurrent, unusual, associated with systemic signs, occurs in an immunocompromised host, or fails initial therapy.

Antibiotics are added to drainage when there is surrounding extensive cellulitis, fever, systemic inflammatory response, hypotension, rapidly progressive disease, multiple lesions, extremes of age, immunosuppression, difficult-to-drain anatomy, indwelling device, or poor response to drainage alone. Empiric MRSA-active therapy may be reasonable when local MRSA prevalence is high or the patient has a prior MRSA history.

Necrotizing fasciitis and necrotizing soft tissue infection

Necrotizing fasciitis is a limb- and life-threatening infection that tracks along fascial planes and can destroy tissue faster than surface findings suggest. Early disease may have only swelling, tenderness, and fever. The clue is disproportionate pain, rapid progression, systemic toxicity, or exam findings that do not match a simple cellulitis. Later findings include skin anesthesia from nerve destruction, bullae, ecchymosis, crepitus, necrosis, severe edema beyond erythema, septic shock, renal injury, and metabolic acidosis. Risk factors include diabetes, recent surgery, trauma, injection drug use, immunosuppression, peripheral vascular disease, and water or soil exposure.

Management is not antibiotic-only. Immediate surgical exploration and debridement are the decisive steps; antibiotics are adjunctive. Empiric therapy should cover MRSA, streptococci, gram-negative rods, and anaerobes. Clindamycin is often included when group A *Streptococcus* or clostridial toxin-mediated infection is suspected because toxin suppression matters. Imaging can help in stable patients when the diagnosis is uncertain, but imaging must not delay surgery in a toxic or rapidly deteriorating patient.

Necrotizing infection clues to memorize:

- Pain out of proportion to visible skin findings.
- Rapid progression over hours rather than days.
- Bullae, ecchymosis, skin anesthesia, or crepitus.
- Toxic appearance, hypotension, altered mental status, AKI, high lactate, or severe metabolic derangement.
- Worsening despite appropriate cellulitis therapy.
- Recent surgery, trauma, diabetes, injection drug use, water exposure, or immunocompromise.

Exposure-based microbiology

Freshwater wounds raise concern for *Aeromonas hydrophila* plus usual skin flora. Severe infections may need gram-negative water exposure coverage.

Saltwater wounds or raw oyster exposure raise concern for *Vibrio vulnificus*, especially in liver disease or iron overload. Rapidly progressive sepsis and necrosis can occur.

Soil, farm injury, crush wounds, and devitalized tissue raise concern for *Clostridium* species and mixed anaerobes. Debridement and tetanus prophylaxis are essential.

Human bite or clenched-fist injury raises concern for *Eikenella*, oral anaerobes, streptococci, and *S. aureus*. The danger is missing tendon sheath or joint capsule involvement.

Animal bites raise concern for *Pasteurella*, *Capnocytophaga*, anaerobes, streptococci, and staphylococci. Severe *Capnocytophaga* infection is especially important in asplenia, alcohol use disorder, and immunocompromise.

Diabetic foot ulcers begin with gram-positive organisms in mild early infection but become polymicrobial with chronicity, depth, ischemia, necrosis, prior antibiotics, or hospitalization.

Lymphadenitis and lymphangitis

Lymphadenitis is inflammation of one or more lymph nodes, often due to drainage from a local skin or soft tissue infection. Lymphangitis produces tender red streaks moving proximally from the wound toward regional lymph nodes. Common pathogens are streptococci and staphylococci, but the differential broadens with cat exposure, animal bites, mycobacterial infection, tularemia, plague, sexually transmitted infections, and malignancy. The source section emphasizes fever, tender regional lymphadenopathy, streaking, and possible progression to sepsis if untreated.

Management includes identifying and treating the primary skin focus, obtaining cultures if there is drainage or systemic illness, and providing antibiotics directed at likely skin flora. Fluctuant nodes or abscesses may require drainage. Persistent, hard, fixed, supraclavicular, generalized, or nonresolving lymphadenopathy requires a broader evaluation beyond routine bacterial infection.

Tetanus and wound prophylaxis

Tetanus is caused by toxin-producing *Clostridium tetani* spores contaminating wounds. The toxin blocks inhibitory neurotransmitter release and causes hypertonicity, trismus, spasms, autonomic instability, and respiratory compromise. It is a toxin disease rather than a spreading cellulitis. High-risk wounds include puncture wounds, avulsions, crush injuries, burns, frostbite, devitalized tissue, wounds contaminated with soil, feces, or saliva, and wounds in patients with incomplete immunization.

Prevention depends on wound cleaning and debridement plus vaccination history. Tetanus toxoid-containing vaccine stimulates active immunity, while tetanus immune globulin provides passive neutralization of unbound toxin in selected high-risk situations. Antibiotics are used for contaminated or infected wounds and for clinical tetanus management, but vaccination and TIG decisions should not be replaced by antibiotics.

Routine wound prophylaxis logic:

- Unknown or fewer than 3 tetanus doses with a clean minor wound: give tetanus vaccine; TIG is generally not needed.
- Unknown or fewer than 3 tetanus doses with a dirty or major wound: give tetanus vaccine plus TIG.
- Completed primary series with booster less than 5 years ago: no booster for clean or dirty wounds.

- Completed primary series with booster 5 to 10 years ago: no booster for clean minor wounds; booster for dirty or major wounds.
- Completed primary series with booster more than 10 years ago: booster for clean minor wounds and dirty or major wounds.

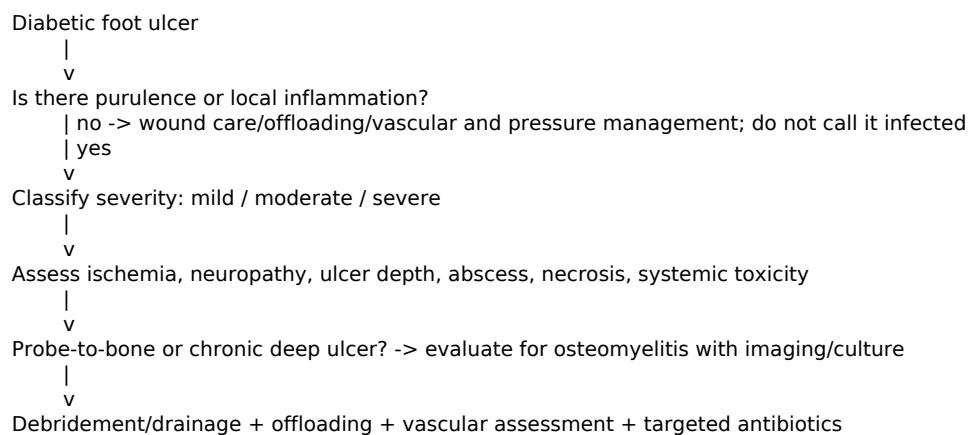
Clinical tetanus is treated in an ICU setting with airway and respiratory support as needed, wound debridement, TIG, active immunization, spasm control such as benzodiazepines, autonomic support, and antimicrobial therapy. Disease does not reliably produce protective immunity, so survivors still require vaccination.

Diabetic foot infection

Diabetic foot infection deserves its own reasoning framework because the visible ulcer is only part of the problem. Neuropathy removes protective pain, ischemia impairs neutrophil delivery and antibiotic penetration, hyperglycemia impairs immunity, and repeated pressure causes deep extension. Not every diabetic foot ulcer is infected: infection requires purulence or at least local inflammatory findings such as erythema, warmth, tenderness, pain, or induration. However, pain and fever may be muted by neuropathy, ischemia, or immune dysfunction.

Classify the infection as mild, moderate, or severe. Mild infection is superficial and limited. Moderate infection is deeper, more extensive, or involves structures such as fascia, tendon, muscle, joint, or bone but lacks systemic toxicity. Severe infection has systemic inflammatory response, metabolic instability, necrotizing features, or sepsis. Evaluate vascular supply, ulcer depth, exposed bone, surrounding cellulitis, abscess, necrosis, foul odor, crepitus, and ability to offload pressure. Probe-to-bone, chronically elevated ESR or CRP, and MRI findings can support osteomyelitis, but bone culture is the microbiologic gold standard when feasible.

Original diabetic foot workflow:



Therapy requires more than antibiotics: debridement of devitalized tissue, drainage of abscess, offloading, glycemic management, vascular evaluation and revascularization when indicated, and wound care. Empiric antibiotics for mild early infection often target gram-positive cocci. Broaden coverage for chronic, previously treated, deep, necrotic, ischemic, foul-smelling, water-exposed, or severe infections. Antibiotic duration depends on tissue depth and source control; diabetes-related osteomyelitis often needs prolonged therapy if infected bone is not resected.

Osteomyelitis

Osteomyelitis is infection-driven inflammatory destruction of bone. It may arise hematogenously, by contiguous spread from an ulcer or soft tissue infection, or by direct inoculation through trauma, surgery, open fracture, injection drug use, or hardware. The source section highlights *S. aureus* as the most common overall organism, with coagulase-negative staphylococci in prosthetic or hardware infection, *Pseudomonas* in puncture wounds through shoes or water exposure, and polymicrobial flora in diabetic foot and decubitus ulcer disease.

Clinical presentation may be subtle. Fever is inconsistent. Localized bone pain is common, and chronic osteomyelitis may form a draining sinus tract. ESR and CRP are nonspecific but useful for monitoring response. Plain films may be normal early because radiographic bone destruction lags behind infection; MRI is generally the most sensitive imaging study for early osteomyelitis and for defining extent. Blood cultures are useful in hematogenous disease. Bone biopsy or operative culture provides the most direct microbiologic diagnosis, especially when empiric therapy might be prolonged or the pathogen is uncertain.

Routes of osteomyelitis

Hematogenous osteomyelitis occurs through bacteremia. In children it often involves long bones; in adults it often involves vertebral bodies. Blood cultures and MRI are high-yield, and clinicians should search for bacteremia sources such as endocarditis, vascular catheter infection, or skin infection.

Contiguous-spread osteomyelitis occurs from diabetic foot ulcers, pressure ulcers, chronic wounds, or adjacent soft tissue infection. It is often polymicrobial, especially when wounds are chronic, deep, ischemic, or previously treated. Probe-to-bone, MRI, debridement, and deep tissue or bone culture are important.

Direct-inoculation osteomyelitis follows open fracture, surgery, penetrating trauma, or hardware placement. The microbiology depends on the exposure. Hardware-associated infections often involve staphylococci and biofilm.

Vertebral osteomyelitis should be considered in persistent focal back pain with fever, bacteremia, dialysis, injection drug use, endocarditis, immunosuppression, or older age. MRI is the key imaging test; biopsy is useful when blood cultures do not identify the organism.

Treatment usually combines prolonged antibiotics with source control. Necrotic bone acts like a foreign body and may require debridement. Hardware infection often requires an orthopedic decision about retention versus removal. Do not reflexively treat every chronic foot ulcer with months of antibiotics; first prove infection, obtain appropriate cultures, debride when needed, and assess vascular supply.

Septic arthritis

Septic arthritis occurs when microorganisms invade the joint space and trigger intense neutrophilic inflammation. It is an emergency because cartilage destruction can occur quickly. The classic presentation is a hot, swollen, painful joint with markedly limited active and passive range of motion. Passive limitation matters because it indicates the joint itself is inflamed; periarticular cellulitis, bursitis, and tendonitis usually preserve passive motion better. Fever may be absent, especially in older or immunocompromised patients.

The most common adult cause is *S. aureus*. Streptococci and gram-negative rods become more important with older age, immune compromise, urinary or GI bacteremia, injection drug use, or healthcare exposure. Gonococcal arthritis should be considered in sexually active young adults and may present with migratory polyarthralgia, tenosynovitis, dermatitis, or purulent monoarthritis. Prosthetic joint infection is a related but distinct syndrome requiring orthopedic management and different diagnostic thresholds.

Diagnostic arthrocentesis is the key step. Send synovial fluid for cell count with differential, Gram stain, culture, and crystals. Crystals do not exclude septic arthritis; gout and infection can coexist. Blood cultures are often positive and should be obtained when systemic illness is present. Empiric antibiotics should follow aspiration when feasible, but they should not be delayed in septic or unstable patients. Drainage is mandatory: repeated needle aspiration, arthroscopic lavage, or open surgical drainage depending on the joint, organism, clinical course, and orthopedic assessment.

Synovial fluid interpretation:

- Noninflammatory fluid is usually clear with low WBC count.
- Crystal arthritis is often inflammatory and may have very high WBC counts.
- Septic arthritis often has WBC count greater than 50,000/microliter with PMN predominance, but lower counts do not exclude infection.
- Gram stain positivity supports infection, but a negative Gram stain does not rule it out.
- Crystals do not exclude infection; keep cultures pending.

Pitfall: do not give intra-articular steroids into an undiagnosed hot joint. Aspirate first, culture first when feasible, and treat urgently if septic arthritis remains plausible.

Prosthetic joint and hardware-associated infection

Prosthetic joint infection can present early after surgery with pain, erythema, drainage, fever, and wound dehiscence; delayed infections may present with chronic pain or loosening rather than fever. Coagulase-negative staphylococci and *S. aureus* are common, and biofilm biology makes eradication difficult without procedural management. Diagnosis uses inflammatory markers, aspiration with synovial studies and culture, and intraoperative cultures when surgery is performed. Treatment strategy depends on timing, implant stability, pathogen, soft tissue condition, and host status: debridement with implant retention, one-stage exchange, two-stage exchange, resection arthroplasty, or suppressive antibiotics may be considered.

High-yield differentials and mimics

Red swollen leg: do not miss necrotizing infection, DVT, abscess, or compartment syndrome. Mimics include venous stasis dermatitis, gout, contact dermatitis, lymphedema, and hematoma.

Painful diabetic foot ulcer: do not miss osteomyelitis, ischemia, abscess, or necrotizing infection. Mimics include neuropathic ulcer without infection and Charcot arthropathy.

Hot monoarthritis: do not miss septic arthritis. Mimics include gout, CPPD, hemarthrosis, reactive arthritis, and Lyme arthritis.

Back pain with fever or bacteremia risk: do not miss vertebral osteomyelitis, epidural abscess, or endocarditis. Mimics include degenerative spine disease, compression fracture, and malignancy.

Facial cellulitis: do not miss orbital cellulitis, cavernous sinus thrombosis, or necrotizing infection. Mimics include preseptal cellulitis, allergic swelling, and sinusitis-related edema.

Initial management algorithms

Algorithm 1: cellulitis or abscess. First ask whether there is a drainable collection. If yes, drain it. Then decide whether antibiotics are needed based on systemic illness, host factors, extent, and MRSA risk. If no purulence, target streptococci and MSSA for mild disease. If severe, rapidly progressive, immunocompromised, or associated with trauma, water, or bite exposure, broaden empirically and reassess for deep infection.

Algorithm 2: suspected necrotizing infection. Recognize red flags, call surgery immediately, obtain blood cultures and labs, begin broad IV antibiotics with toxin-suppressing coverage when appropriate, resuscitate sepsis, and proceed to exploration and debridement. Imaging is a tool for stable uncertainty, not a prerequisite for source control.

Algorithm 3: suspected osteomyelitis. Confirm that symptoms and risk factors match bone infection, obtain ESR or CRP and blood cultures when appropriate, image with MRI when feasible, obtain bone or deep culture if microbiology is uncertain, and combine antibiotics with debridement or hardware planning when necrotic bone, abscess, or foreign material is present.

Algorithm 4: suspected septic arthritis. Aspirate the joint urgently for WBC count with differential, Gram stain, culture, and crystals. Draw blood cultures if systemic illness or bacteremia risk. Start empiric antibiotics after aspiration unless unstable. Arrange drainage and reassess clinically and with inflammatory markers.

Common pitfalls

- Treating an abscess with antibiotics alone and forgetting drainage.
- Using the absence of fever to rule out osteomyelitis, septic arthritis, or diabetic foot infection.
- Waiting for gas on imaging before considering necrotizing fasciitis; gas is specific when present but insensitive.
- Relying on superficial swab cultures from chronic ulcers; they often reflect colonizers rather than the invasive pathogen.
- Giving steroids for presumed gout before aspirating a hot joint when infection is plausible.
- Calling every red swollen leg cellulitis; DVT, venous stasis, gout, and dermatitis are common mimics.
- Forgetting tetanus prophylaxis in wound management.
- Missing vascular insufficiency in diabetic foot infection; antibiotics cannot overcome critical ischemia.
- Assuming a normal early plain radiograph excludes osteomyelitis.

- Failing to reassess within 24 to 48 hours when cellulitis worsens, pain increases, or systemic signs appear.

Practice questions

Question 1. A 62-year-old man with diabetes has a plantar ulcer with surrounding erythema, malodor, and exposed bone. He is afebrile. What is the next reasoning step?

Answer. Do not dismiss infection because fever is absent. Classify severity, assess vascular supply, evaluate bone involvement, obtain appropriate deep tissue or bone cultures when feasible, image with MRI if needed, debride or drain devitalized tissue, and choose antibiotics based on severity and likely organisms.

Question 2. A young adult has a fluctuant forearm abscess without fever or immunocompromise. What intervention matters most?

Answer. Incision and drainage. Antibiotics may be added if there are systemic signs, extensive cellulitis, difficult anatomy, recurrence, immunocompromise, or high MRSA-risk circumstances.

Question 3. A patient with rapidly progressive thigh pain has mild erythema but severe tenderness, tachycardia, hyponatremia, and rising creatinine. What diagnosis must be acted on?

Answer. Necrotizing soft tissue infection. Call surgery immediately, start broad IV antibiotics, resuscitate, and do not delay source control for imaging if unstable.

Question 4. A woman presents with an acutely swollen knee. Synovial fluid has crystals. Can septic arthritis be excluded?

Answer. No. Crystals do not exclude infection. Synovial Gram stain and culture, blood cultures when appropriate, and clinical severity guide urgent treatment and drainage.

Question 5. A puncture wound is contaminated with soil. The patient does not know their vaccine history. What prophylaxis is needed?

Answer. For an unknown or incomplete tetanus immunization history with a dirty or major wound, give tetanus toxoid-containing vaccine and tetanus immune globulin, plus wound cleaning and debridement.

Question 6. An older patient has vertebral back pain, fever, and *S. aureus* bacteremia. What complication should you consider?

Answer. Vertebral osteomyelitis and possibly epidural abscess or endocarditis. MRI spine and blood culture-directed therapy are important; neurologic deficits require urgent evaluation.

Graph-RAG extraction map

Cellulitis

- Class: SSTI syndrome.
- Attributes: nonpurulent, dermis/subcutaneous tissue, streptococcal/MSSA pattern, recurrence with edema or lymphedema.
- Edges: caused by skin barrier disruption; mimics DVT and stasis dermatitis; treated by severity-based antibiotics.

Abscess

- Class: purulent SSTI.
- Attributes: fluctuance, pus, *S. aureus*/MRSA phenotype, culture if severe or recurrent.
- Edges: requires incision and drainage; antibiotics added when systemic or high risk.

Necrotizing fasciitis

- Class: surgical infection emergency.
- Attributes: pain out of proportion, rapid progression, bullae, anesthesia, shock, broad microbiology.
- Edges: requires surgical debridement; associated with diabetes, trauma, injection drug use, and water exposure; toxin suppression may be relevant.

Diabetic foot infection

- Class: complex SSTI and bone infection.
- Attributes: neuropathy, ischemia, ulcer depth, polymicrobial risk, osteomyelitis risk.

- Edges: may progress to osteomyelitis; requires debridement, offloading, vascular assessment, and targeted antibiotics.

Osteomyelitis

- Class: bone infection.
- Attributes: *S. aureus* common, MRI sensitive, bone culture definitive, prolonged therapy.
- Edges: arises via hematogenous, contiguous, or direct inoculation routes; causes chronic sinus tract; interacts with hardware and ischemia.

Septic arthritis

- Class: joint infection emergency.
- Attributes: hot painful joint, limited passive range of motion, synovial WBC/PMNs, culture, drainage.
- Edges: mimics gout; crystals do not exclude infection; *S. aureus* common; gonococcus linked to sexually active young adult pattern.

Tetanus

- Class: toxin-mediated wound infection.
- Attributes: trismus, spasms, autonomic instability, incomplete immunization.
- Edges: prevented by vaccine/TIG decisions; caused by *C. tetani* toxin; requires ICU-level supportive care when clinical disease occurs.

Source control

- Class: management principle.
- Attributes: drainage, debridement, hardware removal or exchange, abscess control.
- Edges: activates in abscess, necrotizing infection, septic arthritis, osteomyelitis, diabetic foot infection, and hardware infection.

Chapter summary

Skin and soft tissue infections are managed by anatomy, purulence, host risk, exposure, and severity. Cellulitis is usually clinical and often streptococcal; abscess requires drainage; necrotizing fasciitis requires immediate surgical action. Diabetic foot infection requires classification, debridement, offloading, vascular assessment, and evaluation for osteomyelitis. Osteomyelitis is a bone culture and MRI problem when diagnosis is uncertain, while septic arthritis is an arthrocentesis and drainage emergency. Tetanus prevention belongs in every wound evaluation because wound appearance alone does not predict immunologic protection.

Selected external references used for clinical cross-checking

- IDSA practice guideline resources for skin and soft tissue infection frameworks and related infectious disease syndromes.
- IWGDF/IDSA 2023 guideline for diabetes-related foot infections, including osteomyelitis duration and remission follow-up concepts.
- SANJO 2023 guideline for management of septic arthritis in native joints.
- CDC clinical guidance for tetanus wound management and tetanus immunization recommendations.
- Uploaded Medvale source coverage: wound and soft tissue infections, cellulitis, erysipelas, necrotizing fasciitis, lymphadenitis, tetanus, osteomyelitis, and acute infectious arthritis.